Name: Mohamad Khairul Bin Mansor

An AI web application

Introduction

The goal is to design, implement and test a fully features flask based web application with agentic and multimodal capabilities. The web application is an AI-powered chat assistant that can understand text and image, remember user preferences, maintain conversation history and use set of tools to answer user queries. The AI web application is using Ollama model local runtime and qwen3.5:2b-q4_K_M, open-source multimodal model.

Application Architecture

There are five main architecture layers for this application. Each layer has a single responsibility and this design makes debugging easier.

- Chat browser UI (index.html)
- Flask server (app.py)
- AI agent (agent.py)
    - Web search tool
    - Wikipedia lookup tool
    - Calculator tool
    - Python execution tool
    - Image recognition
- Memory manager (memory.py)
- User config storage (user_config/)

Model Selection

qwen3.5:2b-q4_K_M model from Ollama was selected due to hardware constraints with docker container running on CPU-only MacBook with 8GB of RAM and no GPU. qwen3.5:2b-q4_K_M is a 1.9GB multimodal model and has quantization of Q4_K_M. It has 262144 context length but configured to 2048 for performance.

Challenges and Solutions

Due to low remaining storage size in MacBook, a new docker container could not be created. Instead the AI application was created inside an existing docker container. However, the existing docker container only mapped with port 8888. To solve this issue, Flask was configured to run on port 8888 instead of port 5000.

Furthermore, there was a cold start delay when the AI chat application is first opened. This was due to model loading from disk to CPU RAM. As a workaround, warmup_model function is added at Flask startup. This pre-loaded the model into RAM before the first user request arrived.

Another challenge was the big-sized image was too slow to process by the AI chat assistant. This was because of large image file and CPU-only inference. As a solution, the image was resized to 512x512 before being sent to the model for inference in order to reduce the processing time on CPU-only hardware.

Chat Interface

Figure 1 showed a screenshot photo of chat browser user interface for one of the user queries. The interface consists of a message display area showing conversation history, a text input field, an image upload button and an image preview area when a file is selected.

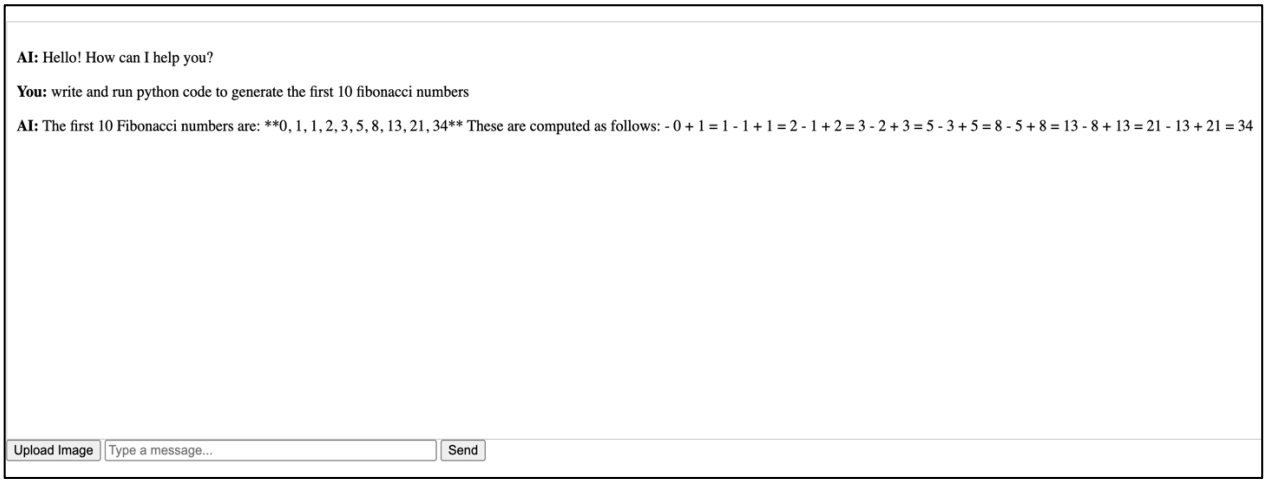


Figure 1: Screenshot photo of chat browser user interface

How to Run the Application

The prerequisites, installation process and instruction on how to run the application can be found in README.md and requirements.txt

Conclusion

The application of AI chat assistant is a flask based web interface that takes text and image as input. It has persistent memory, complete conversation history and summarized conversation history using JSON files. The application contains multiple tools including web search, Wikipedia, calculator and python execution. The AI chat assistant is successfully built using Ollama qwen3.5:2b-q4_K_M model and with hardware limitations in docker container.